

Technical Memorandum, Part 2: Building Footprint Segmentation

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FROM: Sampada Kharel (sampadha.kharel@gmail.com)
DATE: September 22, 2026
RE: Part 2 findings: data partitioning, model/loss design, quantitative results, and visual error analysis (full code: *part2_segmentation.ipynb*)

Data & scope. Inria Aerial Image Labeling dataset: 5000×5000 px, 0.3 m RGB tiles with binary building masks, verified to match the assessment brief exactly. The official test split has no public masks, so **we partition the 180 labeled train tiles ourselves**. For time/compute reasons, a representative **subset of 40 tiles (8 per city × 5 cities)** is used, disclosed explicitly rather than hidden. Building coverage is highly imbalanced and city-dependent: 0.2% (Kitsap) to 46.4% (Vienna), mean 14.4%. A trivial all-background classifier would score over 99% pixel accuracy on the sparsest tiles, motivating the loss and metric choices below.

Leakage-safe partitioning. Each tile is cut into a **non-overlapping 9×9 grid of 512×512 patches** (border discarded); no two patches can share a pixel, by construction, ruling out the overlapping-patch leakage mode by design, not by post-hoc checking. Two splits are used, both verified with executable assertions (no tile spans two sets; every city appears in every split of the primary partition): a **primary split** (tile-disjoint, per-city-proportional: 6 train / 1 val / 1 test tile per city) for the headline result, and a **leave-one-city-out split** (Kitsap held out entirely) as a direct, quantified probe of geographic leakage across cities.

Model & loss. U-Net with a ResNet34 encoder pretrained on ImageNet (skip connections preserve fine edge detail a plain classifier would lose). Loss is **0.4×BCE + 0.4×Dice + 0.2×BoundaryDice**: Dice optimizes overlap and is robust to the class imbalance above (an empty prediction scores Dice ≈ 0); BCE gives a stable gradient early in training; BoundaryDice is a Dice loss on a narrow band around the ground-truth boundary (dilation minus erosion, about 3px wide on each side), targeting edge precision directly in the loss rather than only measuring it afterward. Both models were trained for 30 epochs (AdamW, mixed precision, batch 16, model selection by validation IoU, never by test performance); neither triggered early stopping.

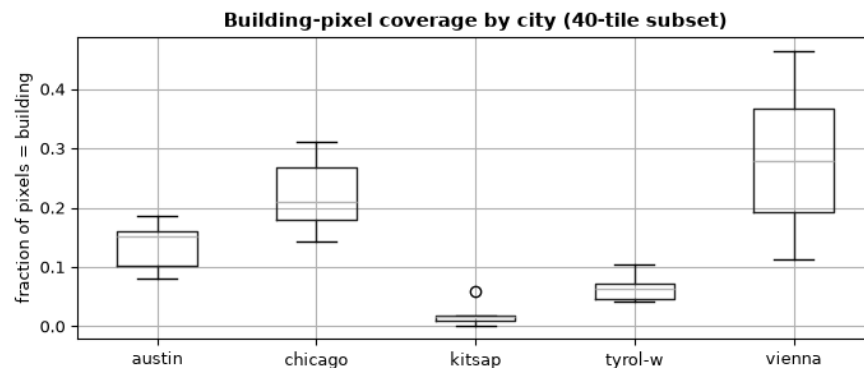


Figure 1. Building-pixel coverage by city (40-tile subset). The imbalance and its city-dependence motivate the loss/metric choices above.

Quantitative results

Both test sets were touched exactly once, using the checkpoint selected by validation IoU.

Eval set	IoU	Dice	Precision	Recall	PixelAcc	BoundaryIoU	Boundary F1
Primary (unseen tiles, known cities)	0.770	0.870	0.848	0.893	94.8%	0.564	0.721
LOCO (Kitsap, city never seen)	0.677	0.808	0.853	0.767	99.4%	0.518	0.683

The geographic-leakage comparison

The same held-out city (Kitsap), scored once by a model that saw *other tiles* of that city during training, and once by a model that never saw the city at all. This is the direct answer to the assessment's geographic-leakage prompt.

Metric	In-distribution (city seen)	Held-out (city never seen)	Gap
IoU	0.816	0.677	0.139
Dice	0.899	0.808	0.091
Precision	0.878	0.853	0.025
Recall	0.921	0.767	0.154

IoU drops 0.139 (17% relative) once the model has never seen any tile from the target city, concrete evidence that city-level generalization is harder than tile-level generalization, and that the per-city-proportional primary split is doing real work, not a formality.

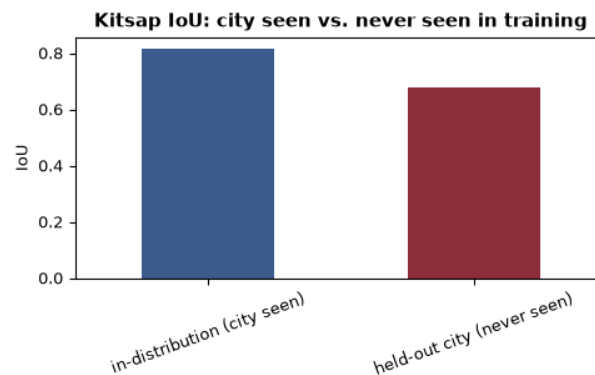


Figure 2. Kitsap IoU: city seen in training vs. entirely held out.

Per-city breakdown (primary split, test set)

City	IoU	Dice	Precision	Recall	PixelAcc	BoundaryIoU
Austin	0.742	0.852	0.846	0.858	95.4%	0.606
Chicago	0.723	0.839	0.793	0.892	90.4%	0.535
Kitsap	0.816	0.899	0.878	0.921	99.6%	0.644
Tyrol-w	0.782	0.878	0.873	0.883	98.9%	0.620
Vienna	0.807	0.893	0.882	0.906	90.0%	0.555

Counter-intuitively, sparse/rural Kitsap has the highest in-distribution IoU and Boundary-IoU of all five cities, not the lowest. Isolated, high-contrast rooftops against forest/field are an easy case despite little training signal. Sparsity is not the same as difficulty once the model has seen a city's visual style at all (contrast with the leave-one-city-out result: the same city drops sharply once that style has never been seen). Chicago and Vienna, the densest cities, are hardest in-distribution, consistent with the failure cases below: visually ambiguous or closely-spaced structures that get merged, missed, or mistaken for buildings.

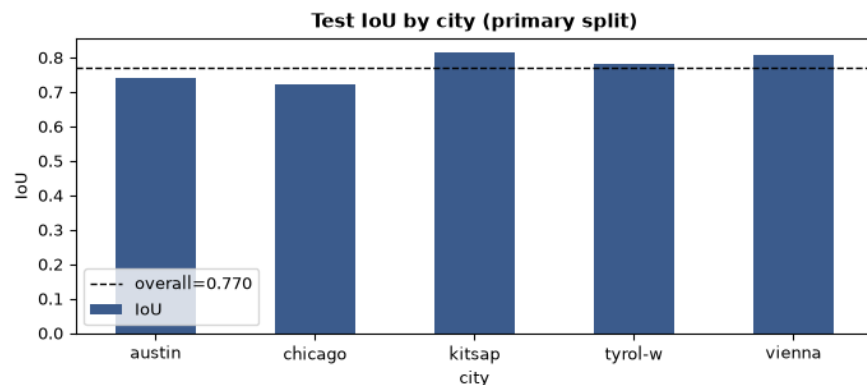


Figure 3. Test IoU by city (primary split); dashed line = overall.

Primary test pixel accuracy (94.8%) looks strong next to IoU (77.0%) purely because background dominates the image. On the sparsest tile in our subset, an all-background prediction alone would already score 99.8% accuracy. IoU, Dice, precision/recall and boundary metrics are what actually distinguish a working model from a lazy one here.

Qualitative visual analysis

Each row: image | ground truth | prediction | error map (green = true positive, red = false positive, blue = false negative). Patch-level IoU selects the examples shown; the metrics that matter are the aggregate ones above.

Best predictions (primary split, test set)

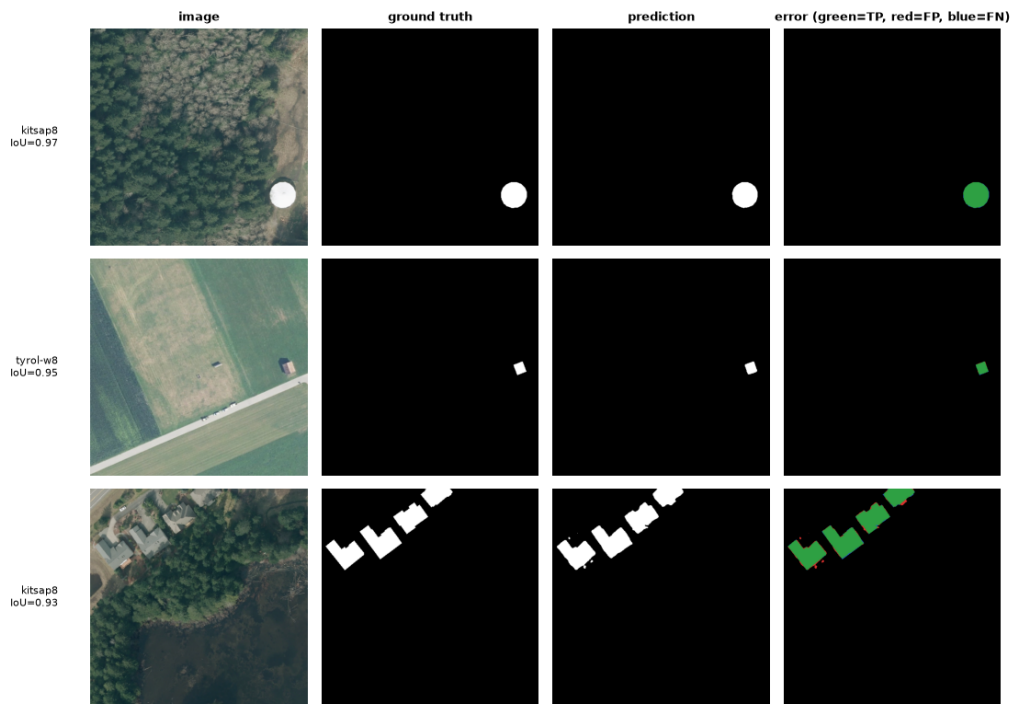


Figure 4. Best predictions (primary split): two Kitsap patches and one Tyrol-w patch, isolated, high-contrast rural buildings are the easiest case for this model.

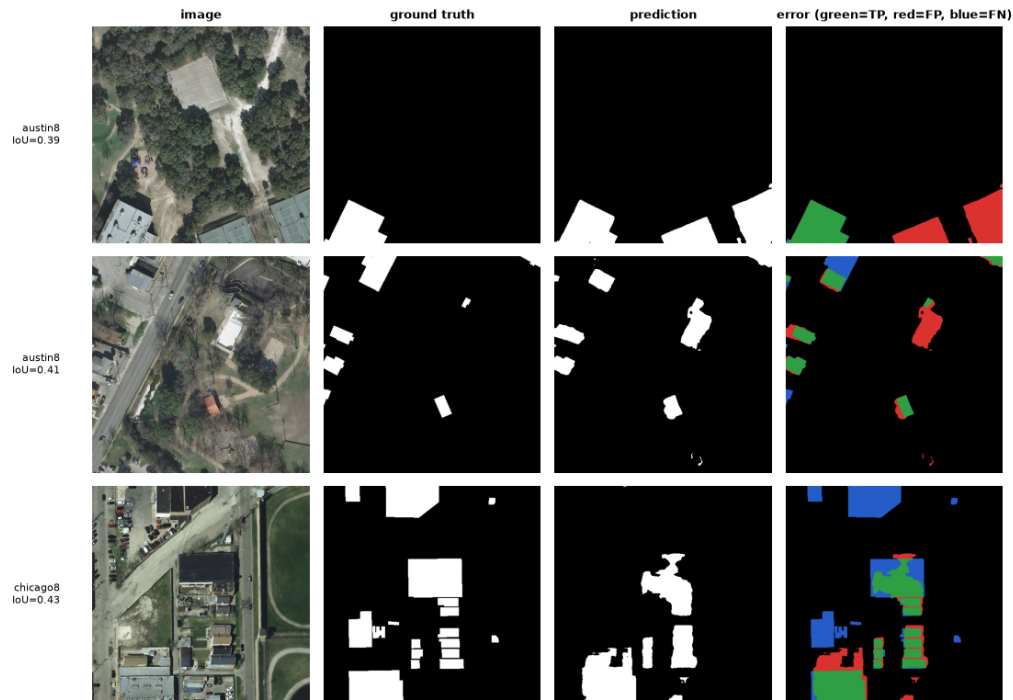


Figure 5. Worst predictions (primary split): Austin and Chicago, but not one single failure pattern. A large false-positive block over a sports-court-like roof, scattered small structures only partially localized, and a dense industrial complex mixing correct detections with over- and under-segmentation.

- **Best cases are isolated rural buildings** (IoU 0.93–0.97, two Kitsap patches and one Tyrol-w patch): a single high-contrast rooftop against a uniform background, with no adjacency or clutter to confuse the model, even though these are the two sparsest cities in the subset.
- **Worst cases (IoU 0.39–0.43, Austin and Chicago) share no single failure pattern:** a large non-building structure mistaken for a rooftop, small structures only partially detected, and a dense complex with mixed correct and incorrect detections. The common thread is visual ambiguity, not one specific building layout.
- **Typical patches** in Austin, Chicago and Vienna are correct at the block level, but scatter errors along the edges between adjacent buildings, a modest instance-adjacency effect expected of semantic (not instance) segmentation, and consistent with these cities' lower Boundary-IoU scores.
- **Geographic unfamiliarity, not sparsity, is what hurts Kitsap:** easiest city in-distribution, but the largest generalization drop once its visual style has never been seen at all.

Limitations & Conclusion

- **Scope:** 40 of 180 labeled tiles (8/city), a compute/time-driven decision, not a claim of matching published full-dataset Inria benchmarks.
- **Leave-one-city-out was run for a single held-out city (Kitsap);** the size of the generalization gap could differ for a different held-out city, though the direction (a drop) would not be expected to reverse.
- **No hyperparameter search:** one reasonable configuration was used for both runs, chosen for time reasons.
- **The boundary-Dice band width (about 3px on each side) was not itself tuned;** other widths, or a distance-transform-based boundary loss, were not compared given time constraints.
- **Instance separation is out of scope** (semantic, not instance, segmentation): adjacent buildings sharing a wall are not required to be told apart, explaining part of the dense-city error pattern above.

Overall, a U-Net/ResNet34 model with a BCE + Dice + boundary-Dice loss reaches a strong, honest validation IoU (~0.78–0.80) on both splits, with no sign of leakage. The boundary term improved Boundary-IoU on every city over a BCE+Dice-only loss. The leave-one-city-out experiment answers the assessment's central geographic-leakage question directly: a 17% relative IoU drop when a city's visual style has never been seen, regardless of how easy that city otherwise is.